Ramanujan Graphs and Zero Forcing in the Generalized Petersen Family

A certified classification of the graphs satisfying the Riemann Hypothesis, and the exact zero forcing number for every large n

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Introduction

The **generalized Petersen graph** $P(n,k)$, $1 \le k < n/2$, has outer vertices u_i and inner vertices v_i ($i \in \mathbb{Z}_n$) with edges u_iu_{i+1}, v_iv_{i+k}, u_iv_i: cubic, on $2n$ vertices, with the rotation $\rho : i \mapsto i+1$ as an automorphism of order n. This board asks two questions about it, one about its fixed adjacency matrix and one about every matrix with its pattern.

What was known. Gera–Stănică [2] computed every eigenvalue of $P(n,k)$ in closed form but never asked which members are Ramanujan; Droll and Le–Sander [6] classified the Ramanujan members of other families (finitely many each). Rashidi et al. [4] proved $Z(P(n,k)) \le 2k+2$ and claimed $Z(P(n,3)) = 8$ for $n \ge 12$; Krishnan [5] exhibited $Z(P(12,3)) = 7$, tabulated $n \le 20$, and conjectured $Z(P(n,3)) = 8$ for all $n \ge 13$, lacking a lower bound valid for every large n. Nothing was known for $k \ge 4$.

Research questions. **Q1.** For which (n,k) is $P(n,k)$ Ramanujan, decided exactly and provably completely?
Q2. Is $Z(P(n,k)) = M(P(n,k)) = 2k+2$ for all large n, and from which n on?

Framework

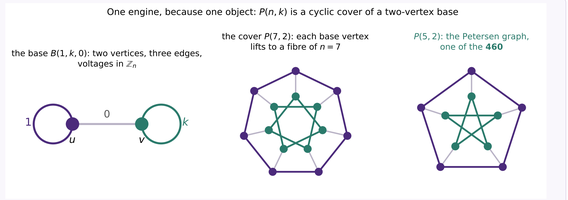


Figure 1. One object seen two ways: the two-vertex base with loop voltages 1 and k, its n-fold cyclic cover $P(n,k)$, and $P(5,2)$, the Petersen graph.

Ramanujan graphs and the graph Riemann Hypothesis. A cubic graph is *Ramanujan* if every eigenvalue other than 3 (and -3 when bipartite) satisfies $|\lambda| \le 2\sqrt{2}$; by Alon–Boppana this is the best spectral gap an infinite family can have. Ihara's zeta function $\zeta_X(u) = \prod_{[C]}(1-u^{\ell(C)})^{-1}$ over prime cycles satisfies $\zeta_X(u)^{-1} = (1-u^2)^{|E|-|V|}\det(I - uA + 2u^2 I)$, so its poles are the roots of $2u^2 - \lambda u + 1 = 0$, one quadratic per eigenvalue λ. With $u = 2^{-s}$ they lie on $\mathrm{Re}\, s = \frac{1}{2}$ iff $|u| = 1/\sqrt{2}$ iff the roots are complex conjugates iff $\lambda^2 \le 8$ [1].

Proposition (Ihara; Terras [1]). A cubic graph satisfies the Riemann Hypothesis for ζ_X if and only if it is Ramanujan. The analogy with the classical RH lives in the zeta function, not in the difficulty: the graph statement is a finite eigenvalue check.

Zero forcing and maximum nullity. Colour a set $S \subseteq V$ blue. If a blue vertex has exactly one white neighbour, that neighbour turns blue. S is a *zero forcing set* if everything ends blue; $Z(G)$ is the least $|S|$. For a symmetric matrix A with $A_{xy} \ne 0 \iff xy \in E$ ($x \ne y$) and free diagonal, $M(G)$ is the largest possible null A.

Proposition (AIM group [3]). $\mathrm{null}\, A \le Z(G)$ for every matrix with the pattern of G; hence $M(G) \le Z(G)$.

Proof sketch. If $Ax = 0$ and x vanishes on the blue set, row u of $Ax = 0$ reduces to $A_{uw}x_w = 0$ for the unique white neighbour w, and $A_{uw} \ne 0$ forces $x_w = 0$. A kernel vector vanishing on a forcing set is 0, so $\ker A$ is determined by its values on S. □
A matrix of nullity $2k+2$ is therefore a *certificate* for a lower bound on Z, checkable by elimination.

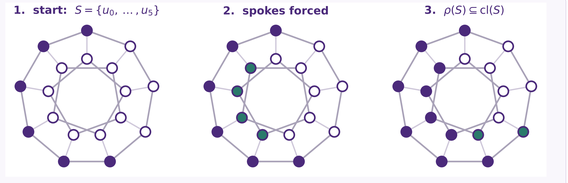


Figure 2. Zero forcing on $P(n,k)$: $2k+2$ consecutive outer vertices force the whole graph, giving $Z \le 2k+2$ [4].

The spectrum on a grid. ρ commutes with A, so the Fourier transform over $\mathbb{Z}_n$ splits A into 2×2 blocks $M_j = \left(\begin{smallmatrix} \alpha_j & 1 \\ 1 & \beta_j \end{smallmatrix}\right)$, $\alpha_j = 2\cos\frac{2\pi j}{n}$, $\beta_j = 2\cos\frac{2\pi jk}{n}$, with eigenvalues $\frac{\alpha+\beta}{2} \pm \sqrt{(\frac{\alpha-\beta}{2})^2 + 1}$: for fixed k, two fixed curves in $t = 2\pi j/n$ sampled at n points. Both halves of the project live on this grid of n-th roots of unity.

Findings

Part I — which $P(n,k)$ satisfy the Riemann Hypothesis

Theorem 1 (polynomial criterion). Let $u = \cos\frac{2\pi j}{n}$ and T_k the Chebyshev polynomial, and put

$$D_k(u) = \big(7 + 4uT_k(u)\big)^2 - 8\big(2u + 2T_k(u)\big)^2 \in \mathbb{Z}[u], \qquad \deg D_k = 2k+2.$$

Then $P(n,k)$ is Ramanujan iff $D_k(u_j) \ge 0$ for every nontrivial j.

Proof sketch. Write $A = \alpha + \beta$, $P = \alpha\beta$. The condition $\lambda_+ \le 2\sqrt{2}$ is $\sqrt{(\alpha-\beta)^2+4} \le 4\sqrt{2} - A$; the right side is $\ge 4\sqrt{2} - 4 > 0$, so squaring is an equivalence, and with $(\alpha-\beta)^2 = A^2 - 4P$ it becomes $2\sqrt{2}A - P \le 7$. The mirror condition gives $2\sqrt{2}|A| \le 7 + P$; since $7 + P \ge 3 > 0$ a second squaring is again an equivalence: $8A^2 \le (7+P)^2$. Substitute $\beta = 2T_k(u)$. □

Theorem 2 (corner criterion). With $x = \cos\frac{\pi j(k+1)}{n}$ and $y = \cos\frac{\pi j(k-1)}{n}$, $P(n,k)$ is Ramanujan iff, at every nontrivial j,

$$Q(x,y) = 4x^2 + 4y^2 - 8\sqrt{2}\,|xy| + 3 \;\ge\; 0.$$

Q does not depend on k, and $\{Q < 0\}$ is four congruent corner pieces covering 0.43% of the square.

Proof sketch. $D_k = f_-f_+$ with $f_\pm = 7 + P \pm 2\sqrt{2}A$ and $f_- + f_+ \ge 6$, so only $\min f_\pm \ge 0$ matters. The identities $\cos t + \cos kt = 2xy$ and $\cos t \cos kt = x^2 + y^2 - 1$ give $A = 4xy$, $P = 4x^2 + 4y^2 - 4$. □

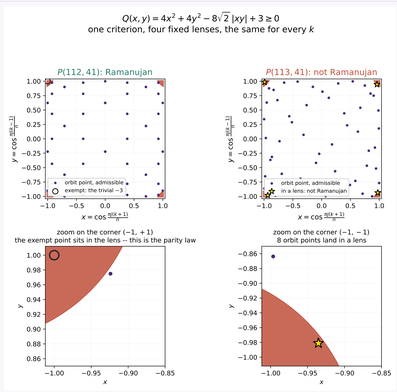


Figure 3. The mapped grid must avoid the same four corners for every k. Left: $P(112,41)$, the largest Ramanujan member; its one point inside a corner is the exempt trivial eigenvalue -3. Right: $P(113,41)$, where a point lands in a corner.

Theorem 3 (finiteness, three ways). (a) $D_k(1) = -7$ for every k, so $n \le B_k = 2\pi/\arccos u_k$, u_k the largest root of D_k below 1; $B_2, B_3, B_4 = 23, 32, 42$, all sharp. (b) Uniformly in k, and (c) for every k at once:

$$n \;\le\; \pi(2+\sqrt{2})\sqrt{k^2+1}, \qquad n \;\ge\; 231 \;\Rightarrow\; \text{not Ramanujan.}$$

(d) Parity: $D_k(-1) = -7$ for odd k, and odd n is then cut at exactly half the constant; even n is exempt, its sample being the trivial -3 of a bipartite graph.

Proof sketch. (b) With $a = 1 - \cos t$, $b = 1 - \cos kt$, $s = a + b$, failure reads $(8\sqrt{2}-4) - (4\sqrt{2}-4)s - 4ab > 7$; AM–GM $4ab \le s^2$ and the identity $s^2 + (4\sqrt{2}-4)s = 8\sqrt{2} - 11$ at $s = \tau := 3 - 2\sqrt{2}$ show $s < \tau$ suffices; $1 - \cos\theta \le \theta^2/2$ gives $s \le (1+k^2)t^2/2$ at $t = 2\pi/n$. (c) Failure follows from $\|j/n\|^2 + \|jk/n\|^2 < \tau/2\pi^2$ for some j. Dirichlet's theorem with $J = \lfloor\sqrt{n}\rfloor$ gives $j \le \sqrt{n}$ with $\|jk/n\| < 1/\sqrt{n}$, so the left side is $< 2/n$, and $2/n < \tau/2\pi^2$ iff $n > 4\pi^2/\tau = 230.1$. □

Theorem 4 (complete classification). Exactly **460** pairs (n,k) satisfy the Riemann Hypothesis, 324 graphs up to isomorphism, out of 13110 cases with $n < 231$. The largest is $n = 112$ ($k = 41$); none has $k > 45$.

The list follows from a structural statement: $P(n,k)$ is Ramanujan iff no divisor m of n is *bad* for k mod m (locality), where badness is witnessed by a single j with $Q < 0$ and $m \le 231$. So the classification is a theorem together with a finite table of bad levels, each entry decided exactly in $\mathbb{Z}[\zeta_m]$. For small k no table is needed:

Theorem 5 (closed form for $k \le 9$). For $k \in \{1,2,3,4,5,7,9\}$ the negative set of D_k on $[-1,1]$ is the end band $(u_k, 1]$ alone (even k) or $[-1,-u_k)\cup(u_k,1]$ (odd k), by exact root isolation of one integer polynomial; hence $P(n,k)$ is Ramanujan iff $2k < n \le B_k$, and for odd k additionally $n \le B_k/2$ when n is odd. For $k = 6, 8$ and $k \ge 10$ an interior band appears and the list has gaps.

Part II — the exact zero forcing number

Theorem 6 (the ceiling). For every matrix A with the pattern of $P(n,k)$, symmetric or not,

$$\mathrm{null}\, A = \dim\ker(T - I) \;\le\; 2k+2, \qquad \text{with equality iff } T = I,$$

T the monodromy of the order-$(2k+2)$ recurrence.

Proof sketch. Row u_i of $Ax = 0$ contains one inner coordinate, c_iy_i with $c_i \ne 0$, so y_i is a three-term expression in x_{i-1}, x_i, x_{i+1}. Substituting into row v_i yields a relation on $x_{i-k-1}, \dots, x_{i+k+1}$ whose extreme coefficients $-e_ib_{i+k}/c_{i+k}$ and $-e'_{i-k}b'_{i-k-1}/c_{i-k}$ are nonzero: a linear recurrence of order $2k+2$. Its solutions on $\mathbb{Z}$ form a $(2k+2)$-space; T is the product of the n step matrices and $\ker A \cong \ker(T-I)$. □

Rotation-invariant certificates. With constant weights $(a,1,c,d,e)$ the blocks are $\left(\begin{smallmatrix} a+s_m & c \\ c & d+et_m \end{smallmatrix}\right)$, $s_m = 2\cos\frac{2\pi m}{n}$, and null A counts grid values that are roots of the symbol

$$F(s) = (a+s)\big(d + e\,t(s)\big) - c^2, \qquad \deg F = k+1.$$

Three roots can be prescribed; for $k = 2$, $c^2 = -(s_1+s_2)(s_1+s_3)(s_2+s_3)$, so antipodal roots are forbidden, which settles $k = 2$ for all $n \ge 9$ except $n = 10$ (done by an integer matrix). Galois orbits of $2\cos\frac{2\pi}{d}$ give $Z(P(n,4)) = 10$ for $60, 70, 90 \mid n$ and $Z(P(n,5)) = 12$ for $24 \mid n$.

Theorem 7 (rational-symbol certificates). A rotation-invariant certificate with rational symbol exists iff n is divisible by one of: $7, 9, 15, 20$ ($k = 2$); $10, 24, 42$ ($k = 3$); $60, 70, 90$ ($k = 4$); $24, 126$ ($k = 5$); none ($k = 6$); 120 ($k = 7$). Proof: a rational symbol has a Galois-closed root set, a union of orbits O_d of total size $k+1$; the product of the Ψ_d must have the $k-2$ forced middle coefficients and $c^2 \ne 0$; finitely many cases, enumerated exactly. Hence no rotation-invariant matrix reaches $2k+2$ for all large n.

Breaking the symmetry: tiles. Fix a *dock* of $k+1$ weight positions; a *tile* carries the dock at both ends and is free inside. Any window of width $2k+2$ sees one tile's interior and dock entries only, so $T = \prod_{\text{tiles}} T_{\text{tile}}$ (Figure 5).

Theorem 8 (tiling). If identity tiles exist for every length in $[L, 2L)$, then $Z(P(n,k)) = M(P(n,k)) = 2k+2$ for every $n \ge L$.

Proof sketch. $n < 2L$: one tile. $n \ge 2L$: peel a tile of length L and recurse. The monodromy of the concatenation is I; apply Theorem 6 and the upper bound. □

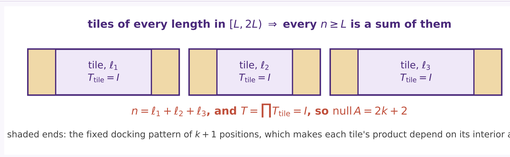


Figure 5. Tiles sharing a dock (amber). A tile of length ℓ has $5(\ell-2k-2)$ unknowns against $(2k+2)^2$ equations, which predicts $\ell \ge 21$ at $k = 3$; that is where tiles first appear.

Existence of the tiles (computer-assisted). $T_{\text{tile}} = I$ iff the one-tile matrix on $P(\ell,k)$ has nullity $2k+2$, i.e. $A(w)K(X) = 0$ with $K = \left(\begin{smallmatrix} I & I \\ X & \end{smallmatrix}\right)$, a bilinear system with integer coefficients. For each required length a solution is shown to exist by Krawczyk's fixed-point theorem, with every bound evaluated in exact rational arithmetic ($\alpha < 1$ as an integer comparison): 77 tiles and 17 single-n instances, $\alpha \le 1.0 \times 10^{-9}$. This is the only step not proved by hand.

Theorem 9 (main result of Part II).

$$Z(P(n,k)) = M(P(n,k)) = 2k+2$$

for all $n \ge 17$ ($k = 3$), $n \ge 29$ ($k = 4$), $n \ge 162$ ($k = 5$), with no congruence condition. A finite case analysis (computer-checked, 1.2×10^9 candidate sets at $P(28,4)$) settles the rest:

n	7	8	9	10	11	12	13	14	15	16	17
$Z(P(n,3))$	6	6	6	8	7	7	8	8	8	8	8
$Z(P(n,4))$	–	–	6	6	7	6	8	8	9	8	9

Corollary. Krishnan's Conjecture 5 holds with the optimal threshold 13; the $k = 4$ row is the first exact determination at any $k \ge 4$.

A withdrawn conjecture. An earlier version of this project conjectured that on a K_4 base the maximum nullity stays 6 while $Z = n+2$, giving $Z - M \to \infty$ on cubic graphs. **This is false:** the three voltage-0 edges form a triangle, and a rank-one block $uu^\top$ on each lifted triangle gives $\mathrm{rank}\, A \le 3n$, so null $A \ge n$. The cover theorem requires $\det M(\zeta) \ne 0$, a hypothesis the earlier statement omitted. The replacement is the leading-coefficient criterion: if the top coefficient of $\det M(\zeta)$ is a monomial in the edge weights, the ceiling holds for every matrix on every cover, by a count on two gradings of the symbol (proved 4 Oct).

Discussion

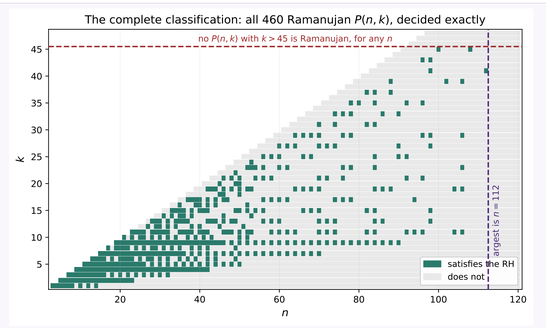


Figure 4. The answer to Q1: all 460 Ramanujan pairs. Unbroken runs where Theorem 5 applies; gaps where an interior band appears; nothing above $k = 45$.

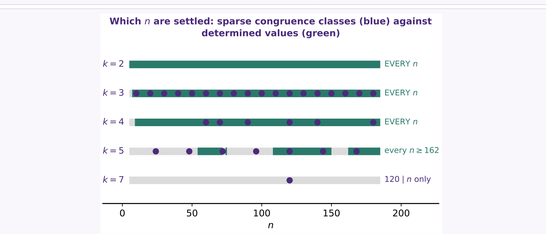


Figure 6. Which n are settled, by k: the sparse congruence classes reached by rotation-invariant certificates (dots) against the values determined for every n by tiles and exhaustive search (solid bars); at $k = 7$ only $120 \mid n$ is known.

Conclusions

- **Q1.** Exactly 460 generalized Petersen graphs satisfy the Riemann Hypothesis. The criterion $Q \ge 0$ contains no k, the bound $n \le 230$ holds for every k, and for $k \le 9$ ($k \ne 6, 8$) the answer is the closed form of Theorem 5.
- **Q2.** $Z(P(n,k)) = M(P(n,k)) = 2k+2$ for all $n \ge 17, 29, 162$ at $k = 3, 4, 5$; Z is known at every n for $k \le 4$; Krishnan's Conjecture 5 is proved with the optimal threshold 13.
- By Theorem 6 the matrix method reaches $2k+2$ or nothing; by Theorem 7 cyclotomic certificates reach only finitely many congruence classes; tiles reach every large n. Every step is proved by hand except tile existence, which is a finite exact computation.
- For a cyclic cover of any fixed base, $M_1 \to M_0$ as $n \to \infty$ and M_0 has top eigenvalue 3, so no infinite Ramanujan family is a cyclic cover; on $P(n,k)$ the deficit $\lambda_2 - 2\sqrt{2}$ stays below $3 - 2\sqrt{2}$.
- Part I is a classification of a known type (cf. Le–Sander [6]); Part II closes two open problems from [4] and [5].

Open problems

- $M(P(24,4))$: $Z = 10$ there, no nullity-10 matrix found, and no rotation-invariant one can exist.
- An effective threshold $L(k)$: identity tiles of every large length now exist for *every* k (an elliptic dock, a first-order root-space count, and a second-order bracket argument in $\mathfrak{sp}(2k+2)$), so $Z = M = 2k+2$ for all large n at every k, but the bound is ineffective and no tile beyond $k = 5$ has been exhibited.
- The cover ceiling when the top coefficient of $\det M(\zeta)$ has two or more monomials, e.g. the tie $t_m = p + q$ on two vertices.
- A proof of tile existence without computation, e.g. an algebraic construction of tiles from two cyclotomic certificates.

References and further reading

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Every number on this board is regenerated from the stored certificates (148 checks) computation in Python (NumPy, SymPy, mpmath).